Take Home Test

Driving Business Growth Through E-Commerce Funnel & Behavior Analysis

Optimizing Customer Journey, Traffic Dynamics, and Revenue Growth

Presented By: Joko Santoso

E-COMMERCE GROWTH

DATA ANALYTICS

BUSINESS INTELLIGENCE

2026.10.25

Data & BI Analyst

JOKO SANTOSO

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ABOUT ME

JOKO SANTOSO

Aspiring Data & BI Analyst

Aspiring Business Intelligence & Data Analyst with 4+ years of experience in Japanese manufacturing environments, leveraging data-driven problem solving to improve production performance and operational reliability. Experienced in building dashboards, performing exploratory data analysis, and generating actionable business insights using Python, SQL, Power BI, Tableau, and Excel.

EDUCATION

2022 - 2026
Bachelor of Japanese Language and Literature
Universitas Ngudi Waluyo

2026 - Present
Data Analytics & Business Intelligence
Bootcamp — Dibimbing.id

EXPERIENCES

Automotive Paint Technician Jun 2022 – Present
Giken Co., Ltd.

Mechanical Engineering Jan 2020 – May 2022
PT Techpack Asia (Albea Group)

SOFTWARE SKILLS

PYTHON
POWER BI
GOOGLE ANALYTICS
EXCEL
TABLEAU
MICROSOFT 365
SQL DATABASE
AUTOCAD
INVENTOR
SketchUp

TECHNICAL SKILLS

Hard Skills : Data & BI | Database & Querying | Data Processing & Programming | Data Analysis | Engineering

Soft Skills : Analytical Thinking | Problem Solving | Continuous Improvement | Professional Teamwork | Excellent Communication

KEY COMPETENCIES & PROJECTS

- Product Analytics & Funnel Analysis (Python, EDA, Cohort Analysis)
- HR Analytics & Attrition Drivers (Tableau, Data Modeling)
- Root Cause Analysis & Continuous Improvement (5S, Kaizen)
- Data Integration & Visualization
- Data Preprocessing, Cleaning, & ETL Pipelines

Executive Summary

- **Dataset :** eCommerce Events History in Electronics Store on Kaggle

- **Dataset Overview:** Analysis of e-commerce platform performance covering **90.7K Views**, **5.38K Carts**, and **3.855 Purchases**.
- **Core Goal:** Identify funnel leakages, optimize marketing/push notification timing, and reduce cart abandonment across high-risk categories.
- **Financial Baseline:** Total Revenue reached **$360.7K** with an Average Order Value (AOV) of **$93.6** and a Conversion Rate of **4.3%**.

DIAGNOSTIC CONTEXT

Business Background & Problem Statement

Background

In a highly competitive e-commerce landscape, driving top-of-funnel traffic (Views) is no longer sufficient if users drop off before completing their purchase.

Problem Statement

1. Funnel Leakage: Severe drop-off occurs between *Product View* and *Add to Cart*. Why are users browsing without taking intent-driven actions?
2. High Cart Abandonment: The overall *Cart Abandonment Rate* stands at 28.3%, with specific product categories spiking above 50%.
3. Unoptimized Timing: When do peak transaction hours occur, and how can we align ad spend and promotional campaigns to maximize ROI?

Key Performance Indicators (KPI Overview)

Core Business Metrics

Total Revenue

$360.7K

Primary top-line financial output generated from electronics store transactions.

Conversion Rate

4.3%

Overall percentage of store visits (Views) successfully converting into completed orders (Purchases).

Cart Abandonment Rate

28.3%

Ratio of added shopping carts that were left unpurchased by users.

Average Order Value (AOV)

$93.6

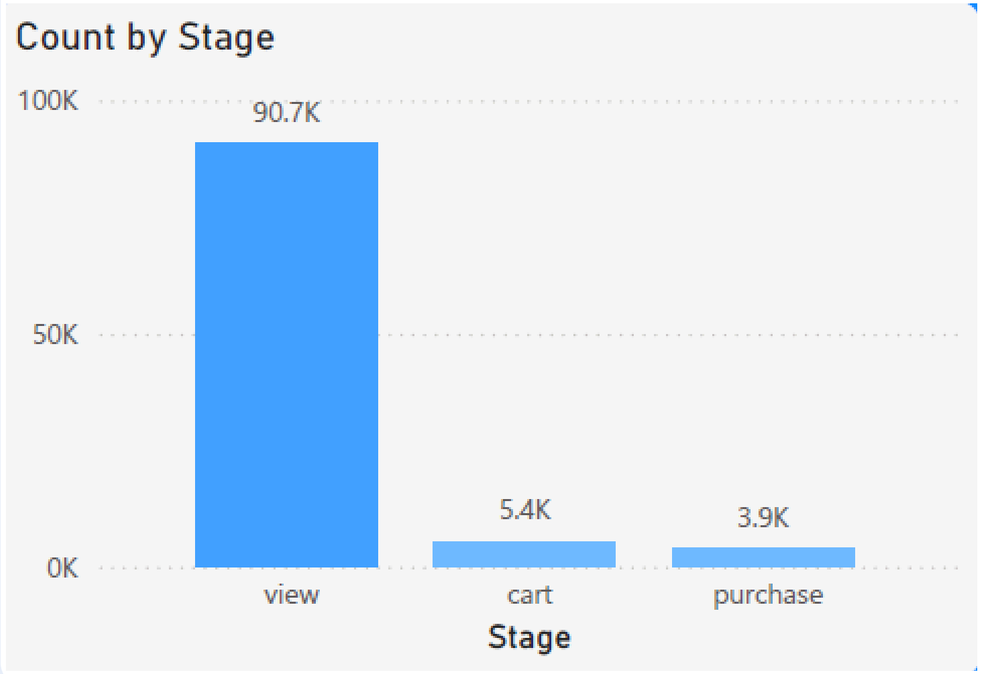
Average monetary spend per completed transaction.

Conversion Funnel Analysis

Views: 91K (100%) → Carts: 5.4K (~5.9%) → Purchases: 3.9K (4.3% Total Conv)

4.4%

Total conversion rate

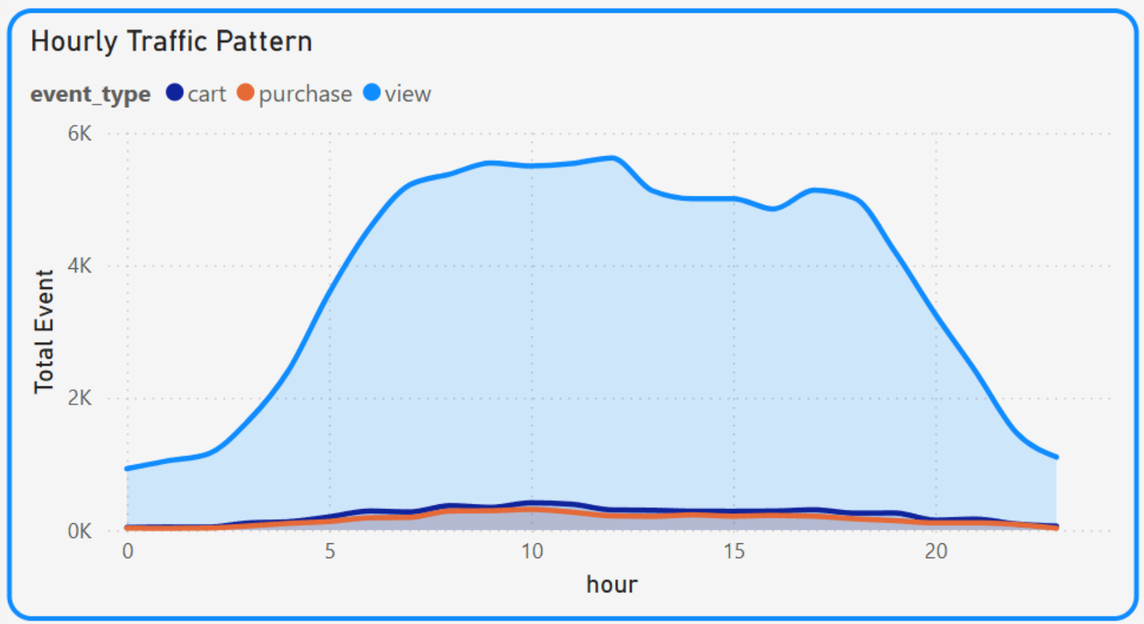


Key Insights

- **Major Leakage Point:** The steepest drop-off happens at the **View** → **Cart** stage. Out of 91K Views, only ~5.9% move to Cart.
- **High Intent Completion:** Once a user adds an item to their cart (*Cart* → *Purchase*), the completion rate is remarkably high (3.8K purchases out of 5.3K carts).
- **Strategic Takeaway:** Optimization efforts should target **Product Detail Pages (PDP)** and add-to-cart incentives rather than the checkout sequence itself.

DIURNAL ENGAGEMENT CURVE

Traffic Dynamics & Peak Hours



Hourly distribution highlights peak platform volume from 10:00 AM to 04:00 PM.

User Behavior Insights

- **Ramp-Up & Peak Hours:** Platform traffic begins climbing at **05:00 AM** and sustains its highest volume between **10:00 AM and 04:00 PM**.
- **Evening Taper:** User engagement gradually decreases after **06:00 PM**, reaching its trough around midnight.
- **Event Correlation:** View events drive the overall traffic curve, while Cart and Purchase trends remain steady and proportional underneath.

Product & Brand Performance

1. Revenue-Driving Categories

- **Computers & Components** and **Peripherals** generate the largest share of total store revenue.
- Computer components alone account for up to **$60.0K** in gross revenue.

$60.0K COMPONENTS REVENUE

2. Brand Concentration

- A substantial share (65.7%) of sales originates from **Unbranded / Unknown** items.
- Among top branded products, market share is led by **AMD** (10.7%), **Canon** (9.3%), ASUS (7.8%), and **Sirius** (6.6%).

65.7% UNBRANDED SHARE

CART FRICTION

Deep-Dive: High Cart Abandonment Categories

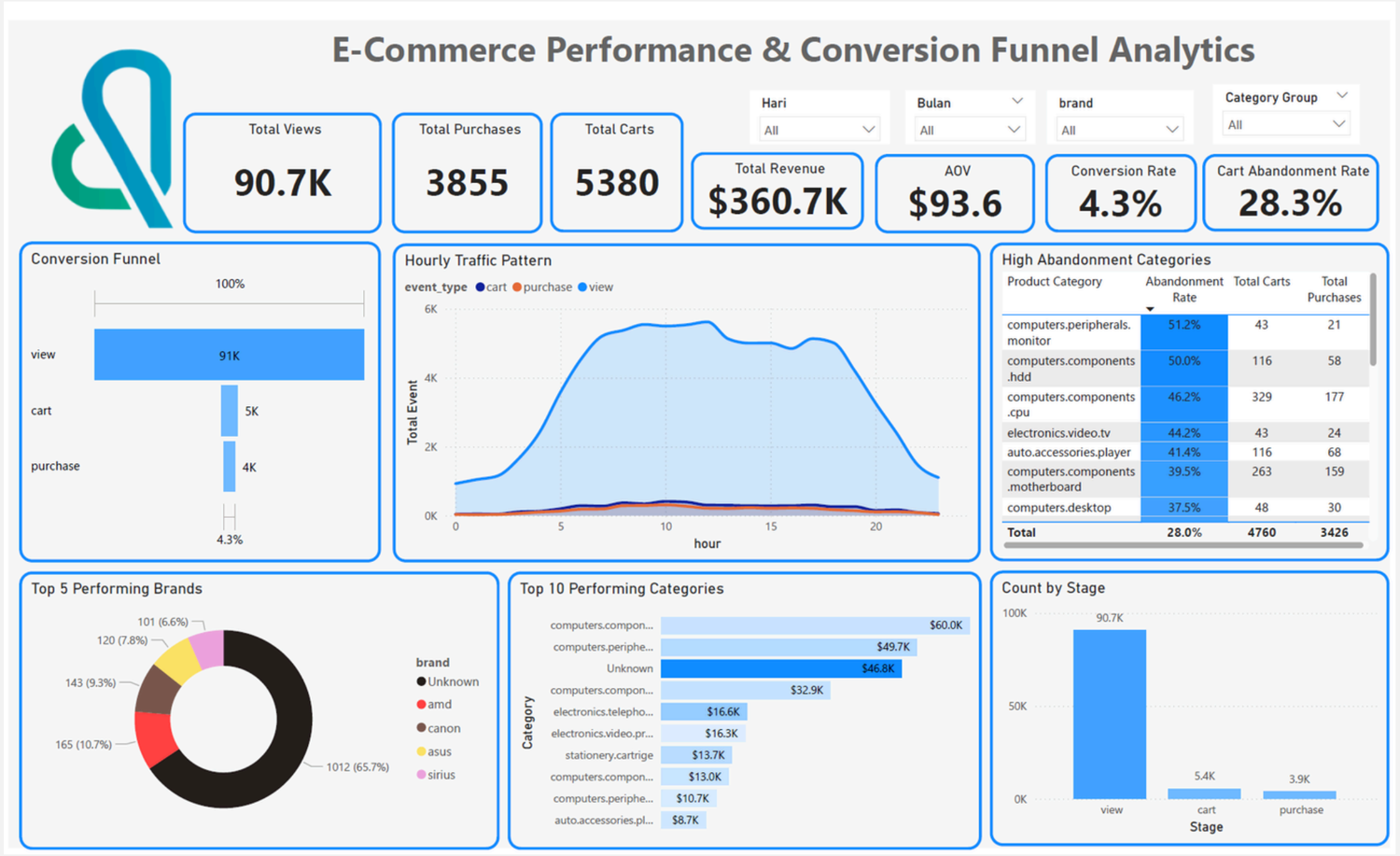
Diagnosis

High-ticket items like **Monitors, HDDs, and CPUs** suffer the highest abandonment rates (46% – 51%). Customers frequently treat their carts as a price-comparison or wishlist holding space for high-investment purchases.

Product Category	Abandonment Rate	Total Carts	Total Purchases
Computers / Peripherals / Monitor	51.2%	43	21
Computers / Components / HDD	50.0%	116	58
Computers / Components / CPU	46.2%	329	177
Electronics / Video TV	44.2%	43	24

Complete Interactive BI Dashboard

Executive Dashboard View



Strategic Business Recommendations

1. Product Page & Funnel Optimization

- Action: Enhance PDPs with high-conversion triggers such as stock scarcity badges ("Only 3 left"), user reviews, and prominent "Buy Now" CTA buttons to streamline View-to-Cart conversions.

2. Automated Abandoned Cart Remarketing

- Action: Launch triggered email and push notification campaigns 2–4 hours after cart abandonment for high-value tech items (Monitors, CPUs, HDDs). Pair notifications with limited-time free shipping or small threshold discounts.

3. Ad Spend & Timing Realignment

- Action: Schedule paid media campaigns (Meta/Google Ads) and promotional push notifications during peak activity hours (09:00 AM – 02:00 PM) to optimize ad budget ROI.

FINANCIAL & OPERATIONAL TARGETS

Conclusion & Expected Impact

Projected Business Impact

+$25K

INCREMENTAL REVENUE

Funnel Conversion Lift: Boosting View-to-Cart conversions by **1.5% – 2%** is projected to yield an estimated **+$15K – $25K** in incremental revenue.

22%

TARGET ABANDONMENT

Reduced Abandonment: Lowering Cart Abandonment from **28.3% down to 22%** via targeted remarketing on tech categories.

15%

AD SPEND SAVED

Marketing Efficiency: Reducing wasted ad spend by up to **15%** through precise time-of-day campaign targeting.

Thank You Very Much!

Any thoughts or reflections?

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